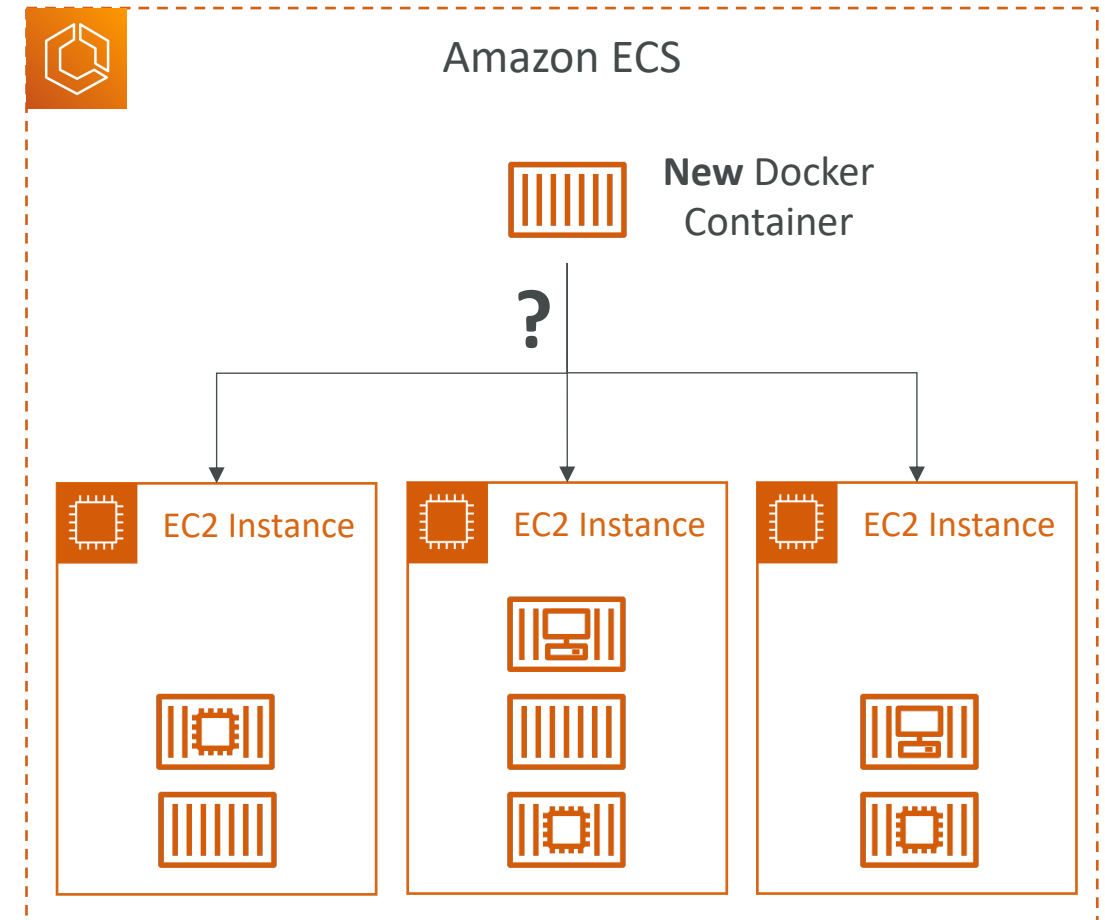


Amazon ECS – Task Placement

- When an ECS task is started with EC2 Launch Type, ECS must determine where to place it, with the constraints of **CPU** and **memory (RAM)**
- Similarly, when a service scales in, ECS needs to determine which task to terminate
- You can define:
 - Task Placement Strategy
 - Task Placement Constraints
- **Note:** only for ECS Tasks with EC2 Launch Type (**Fargate not supported**)



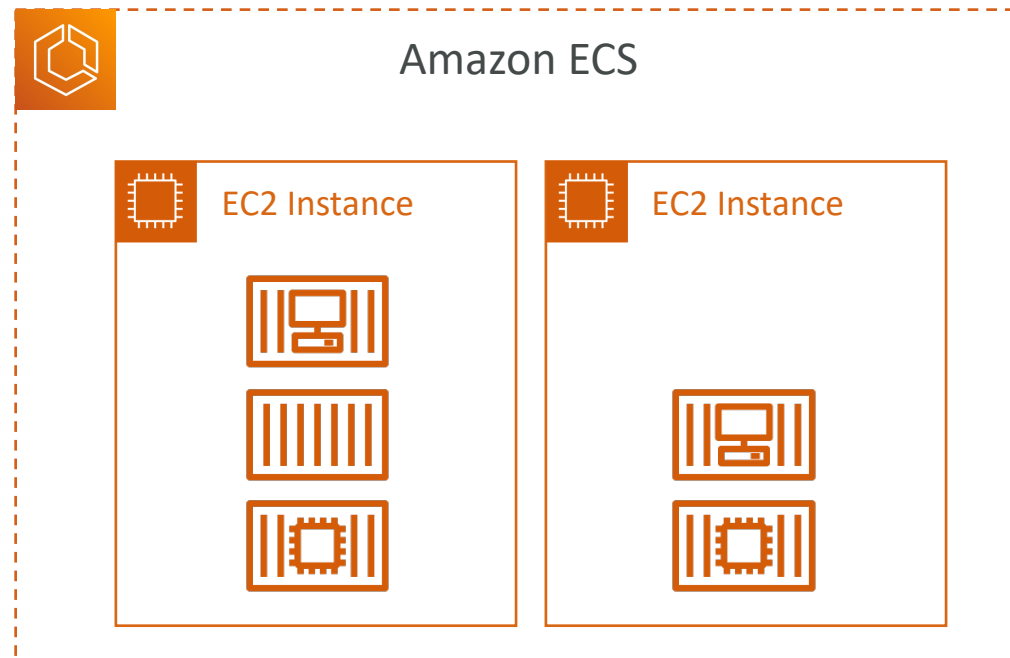
Amazon ECS – Task Placement Process

- Task Placement Strategies are a **best effort**
- When Amazon ECS places a task, it uses the following process to select the appropriate EC2 Container instance:
 1. Identify which instances that satisfy the **CPU, memory, and port** requirements
 2. Identify which instances that satisfy the **Task Placement Constraints**
 3. Identify which instances that satisfy the **Task Placement Strategies**
 4. Select the instances

Amazon ECS – Task Placement Strategies

- Binpack
 - Tasks are placed on the least available amount of CPU and Memory
 - Minimizes the number of EC2 instances in use (cost savings)

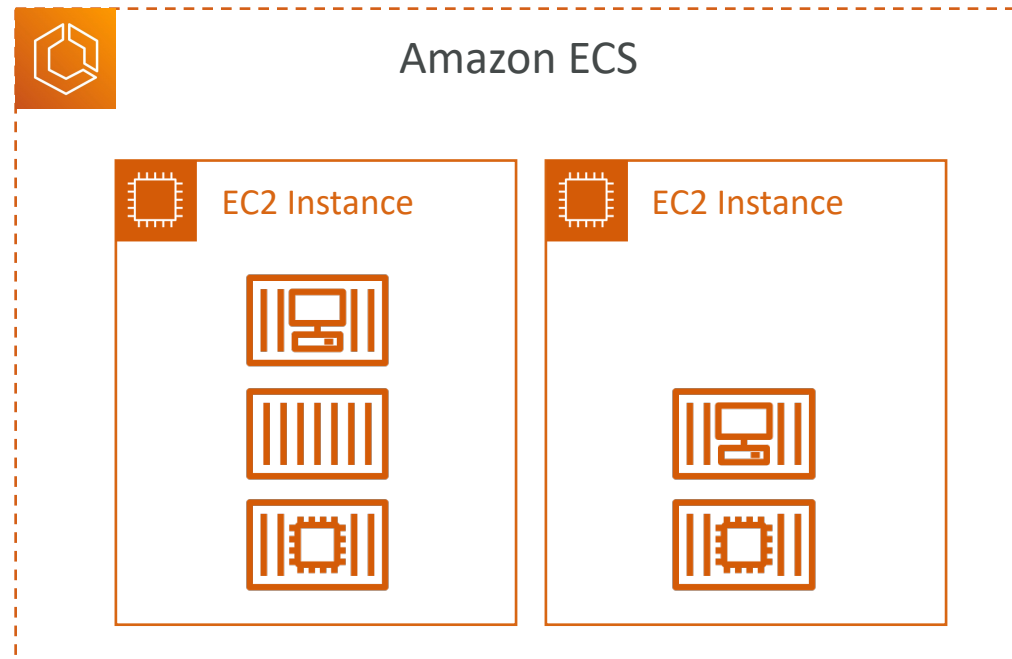
```
"placementStrategy": [  
  {  
    "type": "binpack",  
    "field": "memory"  
  }  
]
```



Amazon ECS – Task Placement Strategies

- Random
 - Tasks are placed randomly

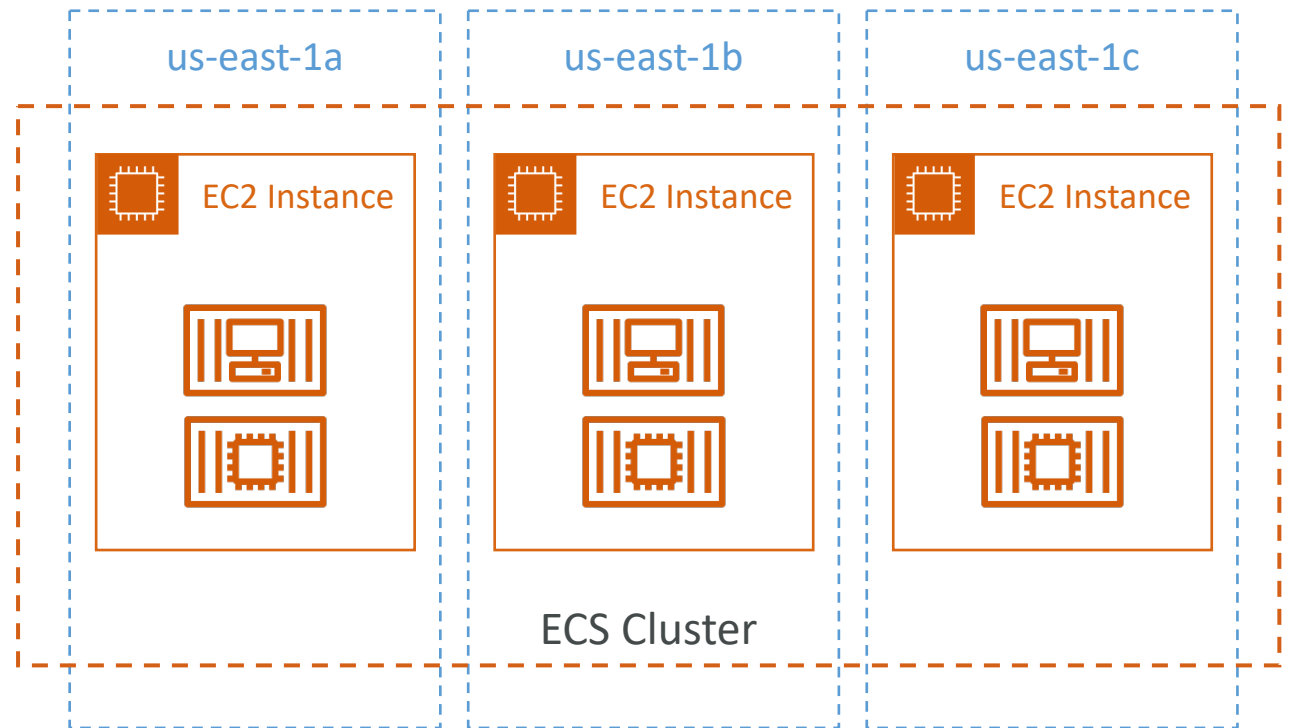
```
"placementStrategy": [  
  {  
    "type": "random"  
  }  
]
```



Amazon ECS – Task Placement Strategies

- Spread
 - Tasks are placed evenly based on the specified value
 - Example: `instancetype`, `attribute:ecs.availability-zone`, ...

```
"placementStrategy": [  
  {  
    "type": "spread",  
    "field": "attribute:ecs.availability-zone"  
  }  
]
```



Amazon ECS – Task Placement Strategies

- You can mix them together

```
"placementStrategy": [  
  {  
    "type": "spread",  
    "field": "attribute:ecs.availability-zone"  
  },  
  {  
    "type": "spread",  
    "field": "instanceId"  
  }  
]
```

```
"placementStrategy": [  
  {  
    "type": "spread",  
    "field": "attribute:ecs.availability-zone"  
  },  
  {  
    "type": "binpack",  
    "field": "memory"  
  }  
]
```

Amazon ECS – Task Placement Constraints

- **distinctInstance**

- Tasks are placed on a different EC2 instance

```
"placementConstraints": [  
  {  
    "type": "distinctInstance"  
  }  
]
```

- **memberOf**

- Tasks are placed on EC2 instances that satisfy a specified expression
- Uses the **Cluster Query Language** (advanced)

```
"placementConstraints": [  
  {  
    "type": "memberOf",  
    "expression": "attribute:ecs.instance-type =~ t2.*"  
  }  
]
```

```
"placementConstraints": [  
  {  
    "type": "memberOf",  
    "expression": "attribute:ecs.availability-zone in  
[eu-west-2a, eu-west-2b]"  
  }  
]
```